# 4.1 Working in an Ordered Field

Section 2.3 proved the rules for inequalities from the order axioms, and Section 2.5 added the one rule that needs division: reciprocals reverse order. A limit argument asks for a few more. It halves a tolerance, so it needs $0 < \varepsilon/2 < \varepsilon$; it subtracts fractions, so it needs $1/b - 1/d = (d-b)/(bd)$; and above all it measures how far apart two numbers are, so it needs the absolute value and the triangle inequality. Each of these is a few lines from the rules we have. This section proves them once, in an arbitrary ordered field, and the rest of the chapter uses them on nearly every page.

## 4.1.1 The standing setup

Throughout this chapter, $F$ is an **ordered field** (Definition 2.3.2): a field with a relation $<$ satisfying (O1)–(O4). The rationals are one, by Theorem 2.5.1, and for most of the chapter they are the one that matters: the next chapter applies these results first with $F = \mathbb{Q}$, to build the real numbers from Cauchy sequences of rationals, and then again with $F = \mathbb{R}$, once $\mathbb{R}$ is shown to be an ordered field. We state results for a general $F$ because the proofs never use anything about $\mathbb{Q}$ beyond (F1)–(F9) and (O1)–(O4), and saying so on the page makes it checkable.

We use the results of Sections 2.3 and 2.5 by their letters, without restating them: the algebra of a commutative ring (Proposition 2.3.2), the rules for inequalities (Proposition 2.3.5 (a)–(i)), and the reversal of order by reciprocals (Proposition 2.5.2). In a field $1 \neq 0$, so Proposition 2.3.5 (e) gives $1 > 0$ in every ordered field. We write

$$F_{>0} := \{x \in F : x > 0\}, \qquad F_{\geq 0} := \{x \in F : x \geq 0\},$$

so that $\mathbb{Q}_{>0}$ of Section 2.5 is the case $F = \mathbb{Q}$. Recall that $a/b := ab^{-1}$ for $b \neq 0$, so $1/b = b^{-1}$, and that $2 := 1 + 1$ and $x^2 := xx$ (Proposition 2.3.2 (d)).

## 4.1.2 Fractions

School algebra adds fractions by cross-multiplying and subtracts reciprocals by a common denominator. Both are identities in any field, and the second is how we will compare $1/b_n$ with $1/b$ in Section 4.4. The only point needing care is that a product of non-zero elements is non-zero, so that $bd$ may stand in a denominator.

**Lemma 4.1.1. Fraction Rules in a Field**

Let $F$ be a field (Definition 2.3.1), and let $a, b, c, d \in F$ with $b \neq 0$ and $d \neq 0$.

(a) $b^{-1}$ is the **only** element $y$ with $by = 1$, and $(b^{-1})^{-1} = b$. Moreover $bd \neq 0$ and $(bd)^{-1} = b^{-1}d^{-1}$.

(b) $\dfrac{a}{b} + \dfrac{c}{d} = \dfrac{ad + bc}{bd}$ and $\dfrac{a}{b} \cdot \dfrac{c}{d} = \dfrac{ac}{bd}$.

(c) $\dfrac{1}{b} - \dfrac{1}{d} = \dfrac{d - b}{bd}$.

(d) For **all** $x, y \in F$: if $xy = 0$, then $x = 0$ or $y = 0$.

*Idea.* Everything rests on one move from Section 2.3: to identify an element, show it satisfies the **defining equation** of the element you want and let uniqueness finish. The product $b^{-1}d^{-1}$ satisfies the defining equation of $(bd)^{-1}$; once $bd$ is known to be a legal denominator,

(b) and (c) are bookkeeping with (F5), (F6) and (F9).

*Proof.*

(a) Suppose $by = 1$. Then

$$y = 1 \cdot y = (b^{-1}b)y = b^{-1}(by) = b^{-1} \cdot 1 = b^{-1},$$

by (F7), (F8) with (F6), (F5), the hypothesis, and (F7); this is the uniqueness argument of Proposition 2.3.2 (a), run with multiplication. Next, $b^{-1} \neq 0$: otherwise $1 = bb^{-1} = b \cdot 0 = 0$ by Proposition 2.3.2 (b), against $1 \neq 0$. So $(b^{-1})^{-1}$ exists, and since $b^{-1}b = bb^{-1} = 1$ by (F6), the element $b$ satisfies its defining equation; uniqueness gives $(b^{-1})^{-1} = b$. Now, by (F5) and (F6),

$$(bd)(b^{-1}d^{-1}) = (bb^{-1})(dd^{-1}) = 1 \cdot 1 = 1.$$

If $bd = 0$, the left side would be $0 \cdot (b^{-1}d^{-1}) = 0$ by Proposition 2.3.2 (b) and (F6), so $1 = 0$, which is false in a field. Hence $bd \neq 0$, and the display says that $b^{-1}d^{-1}$ satisfies the defining equation of $(bd)^{-1}$; by uniqueness, $(bd)^{-1} = b^{-1}d^{-1}$.

(b) By (a), $bd \neq 0$, so both right-hand sides are defined. For the product, by (F5), (F6) and (a),

$$\begin{aligned} \frac{a}{b} \cdot \frac{c}{d} &= (ab^{-1})(cd^{-1}) = (ac)(b^{-1}d^{-1}) \\ &= (ac)(bd)^{-1} = \frac{ac}{bd}. \end{aligned}$$

For the sum, first $\dfrac{ad}{bd} = (ad)(b^{-1}d^{-1}) = (ab^{-1})(dd^{-1}) = \dfrac{a}{b}$, and likewise $\dfrac{bc}{bd} = \dfrac{c}{d}$. Then (F9) with (F6) gives

$$\begin{aligned} \frac{a}{b} + \frac{c}{d} &= (ad)(bd)^{-1} + (bc)(bd)^{-1} \\ &= (ad + bc)(bd)^{-1} = \frac{ad + bc}{bd}. \end{aligned}$$

(c) Apply the sum rule of (b) with $a = 1$ and $c = -1$. Since $(-1)d^{-1} = -(d^{-1})$ by Proposition 2.3.2 (c), the left side is $1/b + (-1)/d = 1/b - 1/d$, and the numerator on the right is $1 \cdot d + b(-1) = d - b$, again by Proposition 2.3.2 (c).

(d) Let $xy = 0$, and suppose $x \neq 0$. Then $y = 1 \cdot y = (x^{-1}x)y = x^{-1}(xy) = x^{-1} \cdot 0 = 0$, by (F7), (F8), (F5) and Proposition 2.3.2 (b). So $x = 0$ or $y = 0$. This proves the lemma.

■

In an ordered field, (d) is also Proposition 2.3.5 (f). The point of (d) here is that it needs **no** order: a field has no zero divisors. The commutative ring of integers modulo 6, in the warning after Definition 2.3.1, has them ($[2][3] = [0]$ with $[2] \neq [0] \neq [3]$), so by (d) it is not a field.

## 4.1.3 Halving and comparing

The commonest move in a limit proof is to give each of two terms half the tolerance. That needs $\varepsilon/2$ to be positive, smaller than $\varepsilon$, and to add up to $\varepsilon$ with itself. The next proposition collects these facts with two comparisons we will need as often.

**Proposition 4.1.2. Halves and Comparisons in an Ordered Field**

Let $F$ be an ordered field.

(a) $\leq$ is a total order on $F$ (Sets and Functions, Definition 5.3.1), and its strict part is $<$.

(b) $2 := 1 + 1 > 0$. For every $\varepsilon \in F_{>0}$,

$$0 < \frac{\varepsilon}{2} < \varepsilon \qquad \text{and} \qquad \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

For all $a, b \in F$ with $a < b$, $a < \dfrac{a+b}{2} < b$.

(c) For $a, b \in F_{>0}$: if $a \leq b$, then $\dfrac{1}{b} \leq \dfrac{1}{a}$.

(d) For $a, b, c, d \in F$: if $0 \leq a \leq b$ and $0 \leq c \leq d$, then $ac \leq bd$.

*Idea.* Nothing new happens: each part is one or two of the rules of Sections 2.3 and 2.5 in sequence. The only care needed is with $\leq$, which hides an "or": each non-strict hypothesis splits into an equality case and a strict case, and the strict rules apply only to the second.

*Proof.*

(a) This is Proposition 2.3.5 (i).

(b) Since $1 > 0$, (O3) gives $2 = 1 + 1 > 0 + 1 = 1$, and $2 > 0$ by (O2). So $2 \neq 0$, and $2^{-1} > 0$ by Proposition 2.5.2 (a). For $\varepsilon > 0$, $\varepsilon/2 = \varepsilon \cdot 2^{-1} > 0$ by (O4). By (F9) and (F7),

$$\begin{aligned} \frac{\varepsilon}{2} + \frac{\varepsilon}{2} &= \varepsilon \cdot 2^{-1} \cdot 1 + \varepsilon \cdot 2^{-1} \cdot 1 \\ &= (\varepsilon \cdot 2^{-1})(1+1) = \varepsilon(2^{-1} \cdot 2) = \varepsilon, \end{aligned}$$

the third equality by (F5) and (F6). Adding $\varepsilon/2$ to $0 < \varepsilon/2$ by (O3) gives $\varepsilon/2 < \varepsilon/2 + \varepsilon/2 = \varepsilon$. For the midpoint, let $a < b$. By (O3), $a + a < a + b$ and $a + b < b + b$, and $a + a = a(1 + 1) = 2a$, $b + b = 2b$ by (F7), (F9), (F6). Multiplying by $2^{-1} > 0$ (Proposition 2.3.5 (c)) gives $a < (a + b)/2 < b$, since $(2a)2^{-1} = a(2 \cdot 2^{-1}) = a$.

(c) If $a = b$, then $1/b = 1/a$. Otherwise $0 < a < b$, and $1/b < 1/a$ by Proposition 2.5.2 (b). In both cases $1/b \leq 1/a$.

(d) First, $ac \leq bc$: if $a = b$ or $c = 0$, the two sides are equal (for $c = 0$ both are 0 by Proposition 2.3.2 (b)); otherwise $a < b$ and $c > 0$, and $ac < bc$ by Proposition 2.3.5 (c). Next, $bc \leq bd$, by the same argument with the roles of the factors exchanged by (F6): $c \leq d$, and $b \geq 0$ since $0 \leq a \leq b$ (transitivity, part (a)). By transitivity of $\leq$ (part (a)), $ac \leq bd$.

■

Part (b) is used whenever a proof splits a tolerance: the sum of two quantities, each below $\varepsilon/2$, is below $\varepsilon$, by Proposition 2.3.5 (b). The midpoint is the same fact read for two points: between any two elements of $F$ lies a third. Part (d) is how a product of two small, or two bounded, quantities is estimated.

*Warning.* The non-negativity in (d) cannot be dropped. In $\mathbb{Q}$, take $a = c = -2$ and $b = d = 0$: then $a \leq b$ and $c \leq d$, but $ac = 4$ and $bd = 0$, and $4 \leq 0$ is false. Section 2.3 gave a strict version of the same trap (the warning after Proposition 2.3.5). Before multiplying two inequalities, check that both sides are non-negative.

## 4.1.4 Absolute value

To say that $x$ is close to $a$, we want one element of $F$ measuring the gap between them, whichever of the two is larger. In school this is $|x - a|$, and $|x|$ is sometimes written $\sqrt{x^2}$. That formula needs a square-root function, and an ordered field has none in general: no rational squares to 2 (Theorem 2.6.1). The square $x^2$ does have the two square roots $x$ and $-x$, and picking the non-negative one is already a definition by cases. So we define the absolute value by cases, using only the order.

*The absolute value of $x$ is $x$ with its sign made non-negative.*

**Definition 4.1.1. Absolute Value in an Ordered Field**

Let $F$ be an ordered field. For $x \in F$, the **absolute value** of $x$ is

$$|x| \coloneqq \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

For $x, y \in F$, the element $|x - y|$ is the **distance** from $x$ to $y$.

The two cases do not overlap and cover $F$, by trichotomy (O1), so $|x|$ is a well-defined element of $F$. The map $x \mapsto |x|$ is a function $F \to F$.

**Examples.** In $\mathbb{Q}$, $|3/4| = 3/4$ and $|-3/4| = 3/4$, the second because $-3/4 < 0$ (Theorem 2.5.1 (a): $(-3) \cdot 1 < 0 \cdot 4$) and $-(-3/4) = 3/4$ (Proposition 2.3.2 (a)). The distance from $1/3$ to $1/2$ is $|1/3 - 1/2| = |-1/6| = 1/6$. The degenerate case is $|0| = 0$, from the first clause. Part (a) below shows that 0 is the only element with absolute value 0, so the distance from $x$ to $y$ is 0 exactly when $x = y$.

**A non-example by minimal change.** Replace the definition by $x \mapsto x^2$. This map is also non-negative, zero only at 0, and multiplicative, so it passes (a) and the product rule of (b) below. It fails the triangle inequality (e): in $\mathbb{Q}$, with $x = y = 1$, the "size" of $x + y$ would be $2^2 = 4$, while the sizes of $x$ and $y$ add up to $1 + 1 = 2$. A measure of size must not grow faster than its argument, and that is the clause the square breaks.

**Proposition 4.1.3. Rules for the Absolute Value**

Let $F$ be an ordered field, and let $x, y, z, r \in F$.

(a) $|x| \geq 0$, and $|x| = 0$ if and only if $x = 0$.
(b) $|-x| = |x|$, $|xy| = |x||y|$ and $|x|^2 = x^2$; if $x \neq 0$, then $|1/x| = 1/|x|$.
(c) $-|x| \leq x \leq |x|$.
(d) $|x| \leq r$ if and only if $-r \leq x \leq r$; and $|x| < r$ if and only if $-r < x < r$.
(e) *(Triangle inequality.)* $|x + y| \leq |x| + |y|$, and $|x - z| \leq |x - y| + |y - z|$.

(f) *(Reverse triangle inequality.)* $||x| - |y|| \leq |x - y|$.

*Idea.* Parts (a)–(d) are case checks on the sign of $x$ (and of $y$). The triangle inequality then needs **no** cases: by (c), $x$ and $y$ are each trapped between $\pm$ their absolute values; adding the two traps traps $x + y$ between $\pm(|x| + |y|)$, and (d) reads that as the inequality. The reverse triangle inequality is the triangle inequality applied to $x = (x - y) + y$ and to $y = (y - x) + x$.

*Proof.*

(a) If $x \geq 0$, then $|x| = x \geq 0$. If $x < 0$, then $|x| = -x > 0$ by Proposition 2.3.5 (a), since $-0 = 0$. So $|x| \geq 0$ always, and $|x| = 0$ only in the first case with $x = 0$; conversely $|0| = 0$.

(b) *Negation.* If $x > 0$, then $-x < 0$ (Proposition 2.3.5 (a)), so $|-x| = -(-x) = x = |x|$ (Proposition 2.3.2 (a)). If $x = 0$, both sides are 0. If $x < 0$, then $-x > 0$, so $|-x| = -x = |x|$.
*Product.* If $x \geq 0$ and $y \geq 0$, then $xy \geq 0$ (by (O4), or Proposition 2.3.2 (b) if a factor is 0), so $|xy| = xy = |x||y|$. If $x \geq 0$ and $y < 0$, then $-y > 0$, and $x(-y) = -(xy)$ by Proposition 2.3.2 (c), so $|x||y| = x(-y) = -(xy)$; and $-(xy) \geq 0$ by the first case, so $xy \leq 0$ and $|xy| = -(xy)$ (if $xy = 0$, both readings give 0). The case $x < 0 \leq y$ is the same with the roles exchanged. If $x < 0$ and $y < 0$, then $xy = (-x)(-y) > 0$ by Proposition 2.3.2 (c) and (O4), so $|xy| = xy = (-x)(-y) = |x||y|$.
*Square.* $|x|^2$ is $xx$ or $(-x)(-x)$, and both are $x^2$ by Proposition 2.3.2 (c).
*Reciprocal.* Let $x \neq 0$. By the product rule, $|x|\,|1/x| = |x \cdot x^{-1}| = |1| = 1$, and $|x| \neq 0$ by (a). So $|1/x|$ satisfies the defining equation of $|x|^{-1}$, and Lemma 4.1.1 (a) gives $|1/x| = 1/|x|$.

(c) If $x \geq 0$, then $-|x| = -x \leq 0 \leq x = |x|$, using Proposition 2.3.5 (a) for $-x \leq 0$. If $x < 0$, then $-|x| = -(-x) = x$, and $x < 0 < -x = |x|$. In both cases $-|x| \leq x \leq |x|$, by transitivity (Proposition 2.3.5 (i)).

(d) ($\Rightarrow$) Let $|x| \leq r$. By (c), $x \leq |x| \leq r$. By (c) and (b) applied to $-x$, $-x \leq |-x| = |x| \leq r$, and negating reverses the inequality (Proposition 2.3.5 (a), or equality), so $-r \leq x$.
($\Leftarrow$) Let $-r \leq x \leq r$. If $x \geq 0$, then $|x| = x \leq r$. If $x < 0$, then $|x| = -x \leq r$, by negating $-r \leq x$. The strict version is the same argument with $<$ in place of $\leq$ at the steps that involve $r$: for instance $x \leq |x| < r$ gives $x < r$ by Proposition 2.3.5 (i).

(e) By (c), $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$. Adding (Proposition 2.3.5 (b)),

$$-(|x| + |y|) \leq x + y \leq |x| + |y|,$$

where $(-|x|) + (-|y|) = -(|x| + |y|)$ by uniqueness of negatives (Proposition 2.3.2 (a)). By (d) with $r := |x| + |y|$, $|x + y| \leq |x| + |y|$. For the second form, apply the first to $x - z = (x - y) + (y - z)$.

(f) By (e), $|x| = |(x - y) + y| \leq |x - y| + |y|$, so $|x| - |y| \leq |x - y|$ by (O3). Exchanging $x$ and $y$, and using $|y - x| = |-(x - y)| = |x - y|$ from (b), gives $|y| - |x| \leq |x - y|$. So $-|x - y| \leq -(|y| - |x|)$ by Proposition 2.3.5 (a) (or equality), and $-(|y| - |x|) = |x| - |y|$ by uniqueness of negatives (Proposition 2.3.2 (a)), since $(|y| - |x|) + (|x| - |y|) = 0$; that is, $-|x - y| \leq |x| - |y|$. By (d), $||x| - |y|| \leq |x - y|$. This proves the proposition.

■

The form of (d) that the rest of the chapter uses is about a distance. Apply the strict half of (d) to $x - a$ with $r = \varepsilon$, and add $a$ throughout by (O3); the step is reversible, since adding $-a$ (by (O3) again) goes back, and $(x - a) + a = x$:

$$|x - a| < \varepsilon \iff a - \varepsilon < x < a + \varepsilon.$$

In words: $x$ lies within $\varepsilon$ of $a$ exactly when it lies strictly between $a - \varepsilon$ and $a + \varepsilon$. In Section 4.3 the role of $x$ is played by the terms $a_n$ of a sequence and $a$ is a candidate limit.

**Example.** In $\mathbb{Q}$, which $x$ satisfy $|x - 1/2| < 1/4$? By the display above, exactly those with $1/2 - 1/4 < x < 1/2 + 1/4$, that is, $1/4 < x < 3/4$. So $2/3$ qualifies ($1 \cdot 3 = 3 < 8 = 2 \cdot 4$ and $2 \cdot 4 = 8 < 9 = 3 \cdot 3$, by the comparison formula of Theorem 2.5.1 (a)) and $3/4$ does not: the inequality is strict, and $|3/4 - 1/2| = 1/4$.

*Quick check.* Is $|a| - |b| \le |a - b|$ for all $a, b \in F$? Is $|a - b| \le |a| - |b|$ for all $a, b \in F$?

*Solution.* The first: yes. By (f) and (c), $|a| - |b| \le ||a| - |b|| \le |a - b|$. The second: no. In $\mathbb{Q}$, take $a = 0$ and $b = 1$: then $|a - b| = |-1| = 1$, while $|a| - |b| = -1$, and $1 \le -1$ is false.

The triangle inequality is the tool of the whole chapter. In its second form it says that a detour through a third point $y$ cannot shorten the way from $x$ to $z$. Run backwards, it turns two small distances, from $x$ to $y$ and from $y$ to $z$, into one small distance from $x$ to $z$. That is the move of almost every proof in Sections 4.3–4.5.

*Remark.* The same two-case formula appears in *Logic and Proof*, Chapter 1, for the school numbers, and in *Analysis on the Real Line*, Chapter 1, for $\mathbb{R}$. Here it is defined, and its rules are proved, for an arbitrary ordered field, from (F1)–(F9) and (O1)–(O4) alone.

## 4.1.5 Exercises

### 4.1.5.1 A. Check your understanding

**Exercise. A1**

Let $F$ be an ordered field.

(a) Here is a proof that $0 < 1/2 < 1$ in $F$: (1) $1 > 0$; (2) $2 = 1 + 1 > 1 > 0$; (3) $0 < 1/2 < 1/1 = 1$. Name the rule behind each of the three steps.

(b) True or false in every ordered field: "if $a < b$, then $a^2 < b^2$". Give a reason, or a counterexample in $\mathbb{Q}$.

(c) True or false in every ordered field: "if $|a| < |b|$, then $a < b$". Give a reason, or a counterexample in $\mathbb{Q}$.

(d) True or false in every ordered field: "$x^2 + 1 > 0$ for every $x$". Give a reason, or a counterexample in $\mathbb{Q}$.

*Solution.*

(a) (a) Proposition 2.3.5 (e), since $1 \neq 0$ in a field. (2) $1 + 1 > 0 + 1 = 1$ by (O3) (or Proposition 2.3.5 (b), strict), then $2 > 0$ by (O2). (3) Proposition 2.5.2 (b), applied to $0 < 1 < 2$; and $1/1 = 1 \cdot 1^{-1} = 1$, since 1 is its own inverse ($1 \cdot 1 = 1$, and Lemma 4.1.1 (a)).

(b) False. In $\mathbb{Q}$, $-2 < 1$, but $(-2)^2 = 4 > 1 = 1^2$. The true statement needs $a, b \geq 0$: Proposition 2.3.5 (g).

(c) False. In $\mathbb{Q}$, $a = 1$ and $b = -2$ give $|a| = 1 < 2 = |b|$, but $a = 1 > -2 = b$.

(d) True. By Proposition 2.3.5 (e), $x^2 > 0$ if $x \neq 0$, and $x^2 = 0$ if $x = 0$ (Proposition 2.3.2 (b)); so $x^2 \geq 0$. Also $1 > 0$. Hence $x^2 + 1 > 0 + 0 = 0$ by Proposition 2.3.5 (b), strict in the second summand.

#### 4.1.5.2 B. Practice

**Exercise. B1: Two companions of (c) and (d)**

Let $F$ be an ordered field.

(a) Prove that if $0 \leq a < b$ and $0 < c \leq d$, then $ac < bd$.

(b) Prove that for $a, b \in F_{>0}$, $a \leq b$ **if and only if** $1/b \leq 1/a$.

Name the rule behind each step.

*Solution.*

(a) Since $0 \leq a < b$, $b > 0$ by Proposition 2.3.5 (i). First, $ac < bc$: $a < b$ and $c > 0$, so Proposition 2.3.5 (c) applies. Next, $bc \leq bd$: if $c = d$ the two sides are equal; otherwise $c < d$ and $b > 0$, so $cb < db$ by Proposition 2.3.5 (c), that is, $bc < bd$ by (F6). Combining $ac < bc \leq bd$ by Proposition 2.3.5 (i) gives $ac < bd$.

(b) ($\Rightarrow$) is Proposition 4.1.2 (c). ($\Leftarrow$) Let $1/b \leq 1/a$. Both are positive, by Proposition 2.5.2 (a). Applying Proposition 4.1.2 (c) to the positive elements $1/b \leq 1/a$ gives $1/(1/a) \leq 1/(1/b)$, and $1/(1/a) = (a^{-1})^{-1} = a$, $1/(1/b) = b$ by Lemma 4.1.1 (a). So $a \leq b$.

**Exercise. B2: Solving an inequality**

In an ordered field $F$, find all $x$ with $|2x - 3| < 5$, naming the rule behind each step. Here 3, 5, 8 and so on denote the multiples $3 \cdot 1$, $5 \cdot 1$, $8 \cdot 1$, ... of Section 2.3 (defined before Proposition 2.3.5), and $2 = 1 + 1 = 2 \cdot 1$.

*Solution.* We use the sum law of Proposition 2.3.5 (h), $(m+k) \cdot 1 = m \cdot 1 + k \cdot 1$; in particular $5 + 3 = 8$, $2 + 3 = 5$ and $4 + 4 = 8$ hold in $F$ because they hold in $\mathbb{N}$.

By Proposition 4.1.3 (d), strict form, $|2x - 3| < 5$ if and only if

$$-5 < 2x - 3 < 5.$$

Adding 3 throughout is reversible (add $-3$ to go back), so by (O3) this holds if and only

if $-5+3 < 2x < 5+3$. Here $5+3 = 8$, and $-5+3 = -(2+3)+3 = -2$, using $-(2+3) = (-2)+(-3)$ (uniqueness of negatives, Proposition 2.3.2 (a)). So the condition is $-2 < 2x < 8$.

Multiplying by $1/2 > 0$ (Proposition 4.1.2 (b)) is reversible too (multiply by $2 > 0$ to go back), so by Proposition 2.3.5 (c) the condition is equivalent to $(-2)/2 < x < 8/2$. Now $2/2 = 1$, so $(-2)/2 = -(2/2) = -1$ by Proposition 2.3.2 (c). And $8 = 4+4 = 4(1+1)$ by (F7) and (F9), the field product of the elements 4 and 2, so $8/2 = 4(2/2) = 4$ by (F5). Hence

$$|2x-3| < 5 \iff -1 < x < 4.$$

Check at an endpoint: $x = 4$ gives $|8-3| = 5$, not $< 5$, as the strict inequality predicts.

**Exercise. B3: Two more triangle inequalities**

Prove that for all $a, b, c$ in an ordered field,

(a) $|a+b+c| \le |a|+|b|+|c|$;
(b) $\big||a|-|b|\big| \le |a+b|$.

*Solution.*

(a) By Proposition 4.1.3 (e) twice, $|(a+b)+c| \le |a+b|+|c| \le (|a|+|b|)+|c|$, the second step by adding $|c|$ to $|a+b| \le |a|+|b|$ (Proposition 2.3.5 (b)). Conclude by transitivity (Proposition 2.3.5 (i)).

(b) Apply Proposition 4.1.3 (f) to $a$ and $-b$: $\big||a|-|-b|\big| \le |a-(-b)|$. By Proposition 4.1.3 (b), $|-b| = |b|$, and $a-(-b) = a+b$ by Proposition 2.3.2 (a). This is the claim.

#### 4.1.5.3 C. Going deeper

**Exercise. C1: A field that cannot be ordered**

(a) Prove that in an ordered field no $x$ satisfies $x^2 = -1$.
(b) Deduce that if a field contains an element $i$ with $i^2 = -1$, then no relation $<$ makes it an ordered field.

*Solution* (Hint 1). Which sign does a square have, and which sign does $-1$ have?

*Solution* (Hint 2). Show first that $x^2 \ge 0$ for every $x$, splitting on whether $x = 0$.

*Solution.*

(a) Let $F$ be an ordered field and $x \in F$. If $x = 0$, then $x^2 = 0$ (Proposition 2.3.2 (b)); if $x \neq 0$, then $x^2 > 0$ (Proposition 2.3.5 (e)). So $x^2 \ge 0$. On the other hand $1 > 0$ (Proposition 2.3.5 (e), as $1 \neq 0$ in a field), so $-1 < 0$ by Proposition 2.3.5 (a). If

$x^2 = -1$, then $0 \le x^2 = -1 < 0$, so $0 < 0$ by Proposition 2.3.5 (i), against (O1).

(b) If some relation $<$ made the field an ordered field, then (a), applied to that ordered field, would forbid $i^2 = -1$. So no such relation exists.

The complex numbers of *Analytic Preliminaries*, Chapter 1 contain such an $i$, so they form a field that cannot be ordered. Compare Example (A field with two elements that cannot be ordered), where the obstacle was $1 + 1 = 0$; here it is that $-1$ is a square.

**Marking scheme.** (10 points) - 4: shows $x^2 \ge 0$ for every $x$ (cases $x = 0$, $x \ne 0$), or rules out $x = 0$ for the $x$ with $x^2 = -1$ and then shows $x^2 > 0$, giving the rule for each step. - 2: shows $-1 < 0$ from $1 > 0$ (justified by $1 \ne 0$ and Proposition 2.3.5 (e), or by citing the standing remark of Section 4.1) and from negation reversing order. - 2: derives the contradiction with trichotomy (O1) or irreflexivity. - 2: (b) states that any order would make (a) apply, so none exists. Fatal errors (0 for the step): asserting $x^2 > 0$ for all $x$ (false at 0) and then using strictness; asserting $1 > 0$ with no justification (accept $1 \ne 0$ with Proposition 2.3.5 (e), or the standing remark of Section 4.1); using properties of $\mathbb{R}$ or of $\sqrt{\ }$.

**Exercise. C2: Maximum and minimum through the absolute value**

For $a, b$ in an ordered field $F$, let $\max\{a, b\}$ be $b$ if $a \le b$ and $a$ otherwise, and $\min\{a, b\}$ be $a$ if $a \le b$ and $b$ otherwise.

(a) Prove that

$$\max\{a, b\} = \frac{a + b + |a - b|}{2}, \qquad \min\{a, b\} = \frac{a + b - |a - b|}{2}.$$

(b) Deduce that $|x| = \max\{x, -x\}$ for every $x \in F$.

*Solution.*

(a) *Case* $a \le b$. Then $a - b \le 0$ (Proposition 2.3.5 (a), or equality), so $|a - b| = b - a$: by the second clause of Definition 4.1.1 if $a - b < 0$, and because both sides are 0 if $a = b$. Hence

$$\frac{a + b + |a - b|}{2} = \frac{a + b + b - a}{2} = \frac{2b}{2} = b = \max\{a, b\},$$

and likewise $\dfrac{a + b - (b - a)}{2} = \dfrac{2a}{2} = a = \min\{a, b\}$, using $b + b = 2b$ and $(2b)2^{-1} = b$ as in the proof of Proposition 4.1.2 (b).
*Case* $b < a$. Then $a - b > 0$, so $|a - b| = a - b$, and the two formulas give $(2a)/2 = a = \max\{a, b\}$ and $(2b)/2 = b = \min\{a, b\}$.

(b) By (a) with $a = x$ and $b = -x$: $\max\{x, -x\} = (x + (-x) + |2x|)/2 = |2x|/2$. By Proposition 4.1.3 (b), $|2x| = |2||x| = 2|x|$, since $2 > 0$. So $\max\{x, -x\} = |x|$.